Space Data and Software: Two Short Articles

1 Astronomy Data Needs Better Software Too

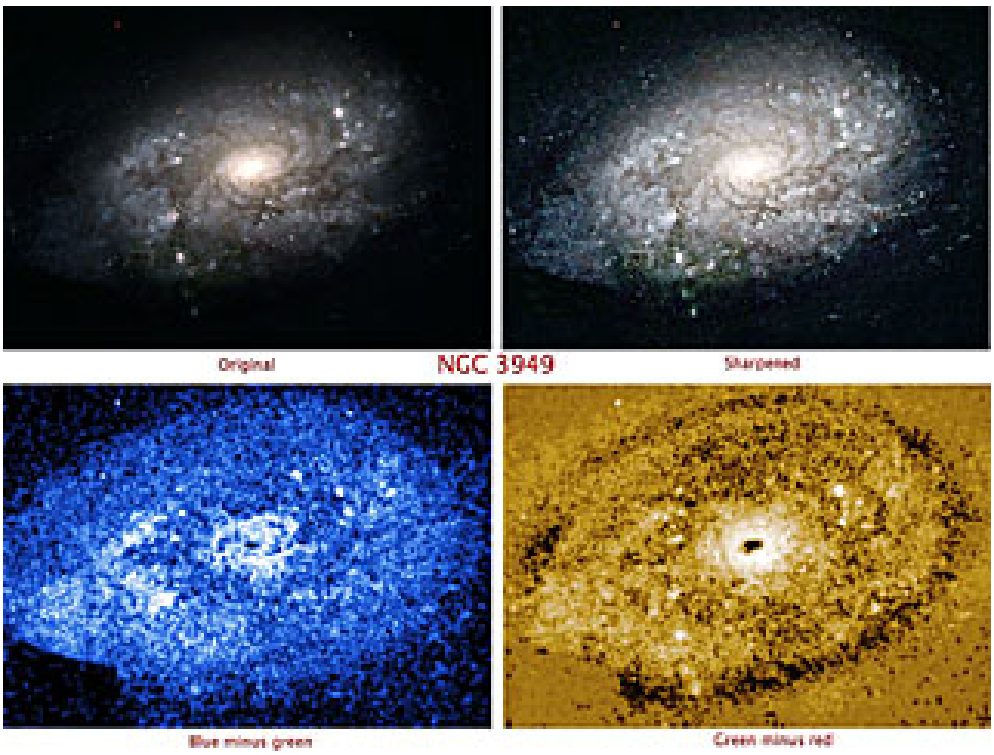


Figure 1: A galaxy shown in its original and processed forms, illustrating how image processing can reveal or emphasise astronomical features. Image credit: Tom Caldwell / ESA-Hubble.

Modern telescopes collect enormous amounts of data from distant stars, galaxies, and other celestial objects. However, collecting this information is only the first step. Scientists also need reliable software to process observations, remove errors, and identify meaningful patterns.

NASA's Artifact InSPECtor citizen-science project helps people identify unwanted signals and image artifacts in telescope data. These corrections can help researchers distinguish genuine astronomical features from distortions.

As telescopes become more powerful, software will play an increasingly important role in astronomy. Better data-processing tools can help scientists analyse observations more accurately and make discoveries that might otherwise remain hidden.

References

- ESA/Hubble, *NGC 3949: original and processed image examples*. https://esahubble.org/projects/fits_liberator/fitsimages/tom_caldwell_1/

- NASA, *Artifact InSPECtor*. https://science.nasa.gov/citizen-science/artifact-inspector/

2 Where Does Space Data Come From?



Figure 2: Artist's rendering of the Landsat 8 satellite observing Earth. Image credit: NASA's Scientific Visualization Studio.

Space data comes from many sources, including space telescopes, planetary missions, astronomical observatories, and scientific instruments. These sources collect information about stars, galaxies, planets, black holes, and other objects in the universe.

For example, the James Webb Space Telescope captures infrared observations of distant galaxies, while planetary missions collect data about the surfaces and atmospheres of other worlds. Earth-observation satellites also collect measurements of land, oceans, clouds, and vegetation. Ground-based observatories contribute by recording light and other signals from space.

The collected data is then processed using specialised software, helping scientists analyse observations, identify patterns, and discover new information about the universe.

References

- NASA Scientific Visualization Studio, *Landsat 8 (LDCM) spacecraft image*. https://svs.gsfc.nasa.gov/10812/
- NASA, *James Webb Space Telescope*. https://science.nasa.gov/mission/webb/
- NASA, *Entry Points to NASA Science Data*. https://science.nasa.gov/learn/entry-points-to-nasa-data/